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$SnSO_4$ modified ZnO nanostructure for highly sensitive and selective formaldehyde detection

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ABSTRACT

$SnSO_4$ modified ZnO nanoparticles have been successfully synthesized for highly sensitive and selective formaldehyde detection after sintering at 450 °C in air. The ZnO/SnO_2 heterojunctions and the surface sulfate play a significantly important role for the excellent formaldehyde sensing performance. Remarkably, the ZnO-3%$SnSO_4$ sensor is able to detect ultralow concentration of formaldehyde (0.06 ppm) with excellent selectivity from commonly occurred indoor VOCs. With engineering efforts, a highly compact prototype formaldehyde sensor is obtained, which is very convenient for portable formaldehyde specific detection.

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1. Introduction

Formaldehyde is a frequently occurred indoor pollutant and was classified as Group I human carcinogen by the International Agency of Research on Cancer in 2004 [1]. The World Health Organization (WHO) has set a very strict 30 min exposure limit of only 0.08 ppm. Presently, the indoor formaldehyde has become a major health threat in developing countries such as China, mainly due to the increasing use of chemical adhesives. Hence, it is highly desirable to monitor the ambient formaldehyde concentration in real time using compact, portable devices instead of complicated instrumentations.

Metal oxide semiconductor (MOS) nanostructures are well known for their gas sensing properties, which is attributed to the change of electric conductivity due to the interaction of the gas molecules and the MOS surface [2–5]. With the advances of modern electronic technologies, MOS nanostructures can be easily integrated into highly compact gas sensing devices to monitor the environmental hazardous gases in real time. However, for most compact MOS sensors, the sensitivity is insufficient to meet the strict WHO standard.

Proper modification of the MOS nanostructures is critical to improve the sensitivity. Controlling the surface electronic states of the MOS nanostructure by surface modification is an important approach to improve the gas sensing properties [6–9]. The morphology of the MOS nanostructures also strongly affects the gas sensing performance [10–19]. Higher specific surface area is beneficial for gas sensing due to higher density of interaction sites with the gas molecules. Meanwhile, the particles in the MOS layer should maintain good electric contact with each other in order to give sensitive conductivity response. As an important criterion for application, the above surface modification and morphology control should be achieve with simple, low cost procedure.

Here, we report a very easy modification of ZnO nanoparticles using $SnSO_4$ to achieve highly sensitive and selective formaldehyde detection. $SnSO_4$ induces connection of ZnO nanoparticles into nanosheets during sintering, accompanied with concomitant formation of the ZnO/SnO_2 heterojunction and the surface sulfate phase. The above structural features amplify the electric conductivity change in different gas environment, resulting in significantly enhanced sensitivity for formaldehyde. With the activation of UV light, ultralow formaldehyde concentration of 0.06 ppm can be detected at room temperature with excellent selectivity. The $SnSO_4$ modified ZnO nanoparticles can be easily prepared using low cost materials, which is highly promising for compact sensing devices for highly sensitive and selective formaldehyde detection.

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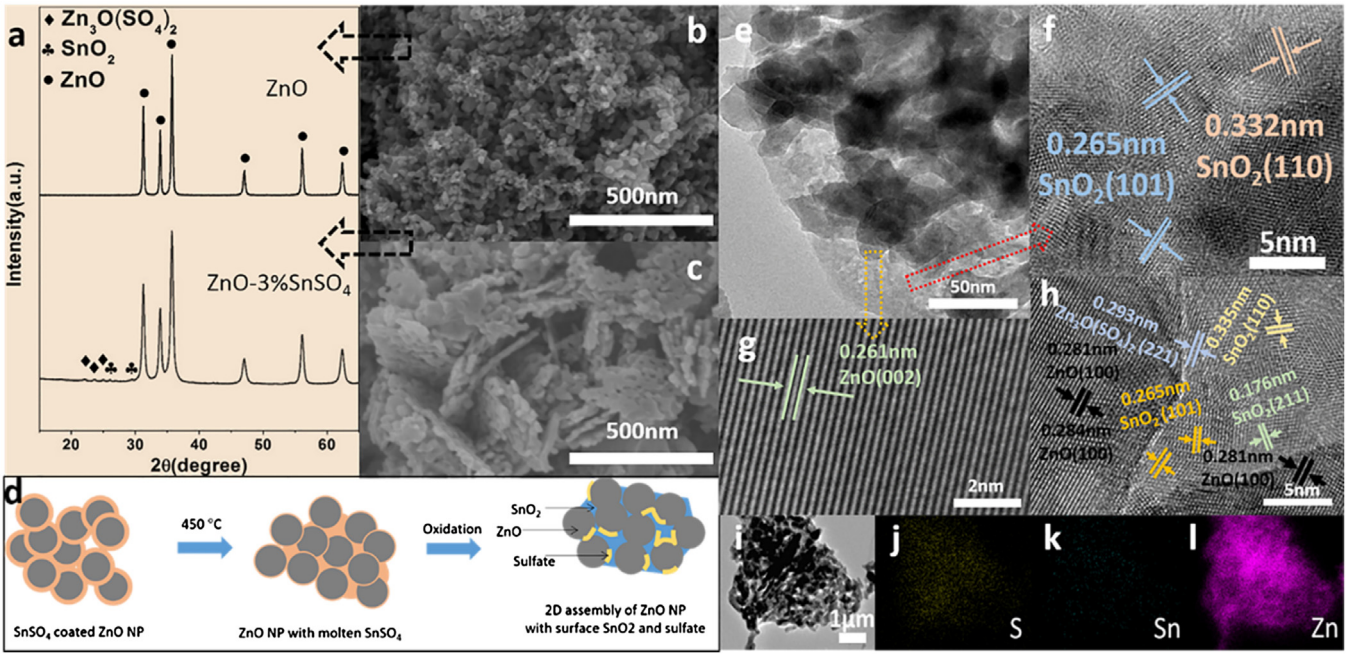


Fig. 1. (a) XRD patterns of pure ZnO nanoparticles and the ZnO-3%$SnSO_4$ sample; SEM images of pure ZnO nanoparticles (b) and the ZnO-3%$SnSO_4$ sample (c); (d) A schematic illustration of the $SnSO_4$ induced 2D assembly of ZnO nanoparticles; (e)–(h) TEM images of the ZnO-3%$SnSO_4$ sample, showing lattice images of the SnO_2 thin layer (f), the ZnO nanoparticle (g) and the ZnO/SnO_2 junction. (i)–(l) EDS mapping images of ZnO-3%$SnSO_4$ sample.

2. Material and methods

2.1. Materials synthesis and characterization

To prepare the ZnO nanoparticles, NH_4HCO_3 solution (50 mL, 1.26 mol/L) is added to $ZnSO_4$ solution (25 mL, 1.5 mol/L) dropwisely under vigorous stirring. The mixture is stirred at 40 °C for 1 h and the precipitate is separated by filtration. The solid product is thoroughly washed with deionized water for three times and is dried in an oven at 120 °C for 4 h. Finally, the dried precursor was calcinated at 500 °C for 2 h to give the ZnO nanoparticles.

The obtained ZnO nanoparticles (0.400 g) are dispersed in aqueous SnSO4 solution (100 mL) with different concentration by sonication. The resulting suspension is heated to 120 °C under vigorous stirring to remove the water. The obtained solid is dried in an oven at 120 °C for 4 h. The sample is calcinated at 450 °C for 4 h to give the $SnSO_4$ modified ZnO samples. The samples are designated as ZnO-x$SnSO_4$, where x is the molar ratio of Sn/Zn.

The nanostructures are characterized by X-ray diffraction (XRD, Rigaku D/max 2000 diffractometer, Cu Kα), scanning electron microscopy (SEM, Hitachi S4800, 10 kV) transmission electron microscopy (TEM, JEOL JEM-2100, 200 kV) with an energy dispersive X-ray spectroscopy (EDS) analyzer and X-ray photoelectron spectroscopy (XPS, AXIS-Ultra, Kratos Analytical, Al Kα).

2.2. Sensor fabrication and gas sensing measurement

The sensor is fabricated according to that reported in our previous work [20]. A printed circuit board (PCB) with two interpolated Au electrode arrays is used as the substrate. The ZnO-x$SnSO_4$ sample is coated onto the substrate by drop casting ethanol suspension of the nanoparticles. After evaporation of the ethanol at room temperature, a semiconductor layer with total material mass loading of 20 mg·cm^{-2} is formed. An UV LED (3 mW) is attached 2 mm above the semiconductor layer to activate the sensing layer during operation.

2.3. Measurement

The gas sensing properties are measured at 30 °C in ambient air. The sensor is sealed inside a glass container with calibrated internal volume about 10 L with a thermostatic water jacket. The current passing through the two Au electrode arrays is measured at a constant voltage of 10 V by an electrochemical workstation (model: CHI 760b). Formaldehyde and other gases with desired concentration are introduced by injecting certain volume of gas or vapor through a septum. The gas inside the glass container can be rapidly refreshed with dry air using a circulation pump with pumping speed of 2 L s^{-1}. The humidity shows negligible influence on the sensor's response and thus is not deliberately controlled, which varies from 20 to 30% during the measurements. The sensor response (R) was defined as

$$R = (i_g - i_0)/i_0 \quad (1)$$

where i_g and i_0 are the current measured under UV irradiation in the presence and absence of the gas molecules, respectively.

3. Results and discussion

The obtained ZnO sample shows pure hexagonal Wurtzite structure (JCPDS NO. 36-1451), as suggested by the XRD pattern (Fig. 1a). For the ZnO-x$SnSO_4$ samples (the samples are designated as ZnO-x$SnSO_4$, indicating $SnSO_4$ modified ZnO, where x is the molar ratio of Sn/Zn), additional diffraction peaks attributed to $Zn_3O(SO_4)_2$ (JCPDS No. 32-1475) and SnO_2 (JCPDS No. 41-1445) also appear in the XRD pattern (Figs. 1a and S1). Interestingly, there is no shift of the diffraction peaks with the increase of $SnSO_4$/ZnO molar ratio, indicating that the $SnSO_4$ modification causes little Sn substitution in the lattice of ZnO (Fig. S1). Thus, the introduced $SnSO_4$ exists as separated $Zn_3O(SO_4)_2$ and SnO_2 phases rather than homogeneously doped into the ZnO lattice, indicating that $SnSO_4$ is oxidized during the sintering in air according to the following reaction:

$$2ZnO + SnSO_4 + O_2 \rightarrow Zn_3O(SO_4)_2 + SnO_2 \quad (2)$$

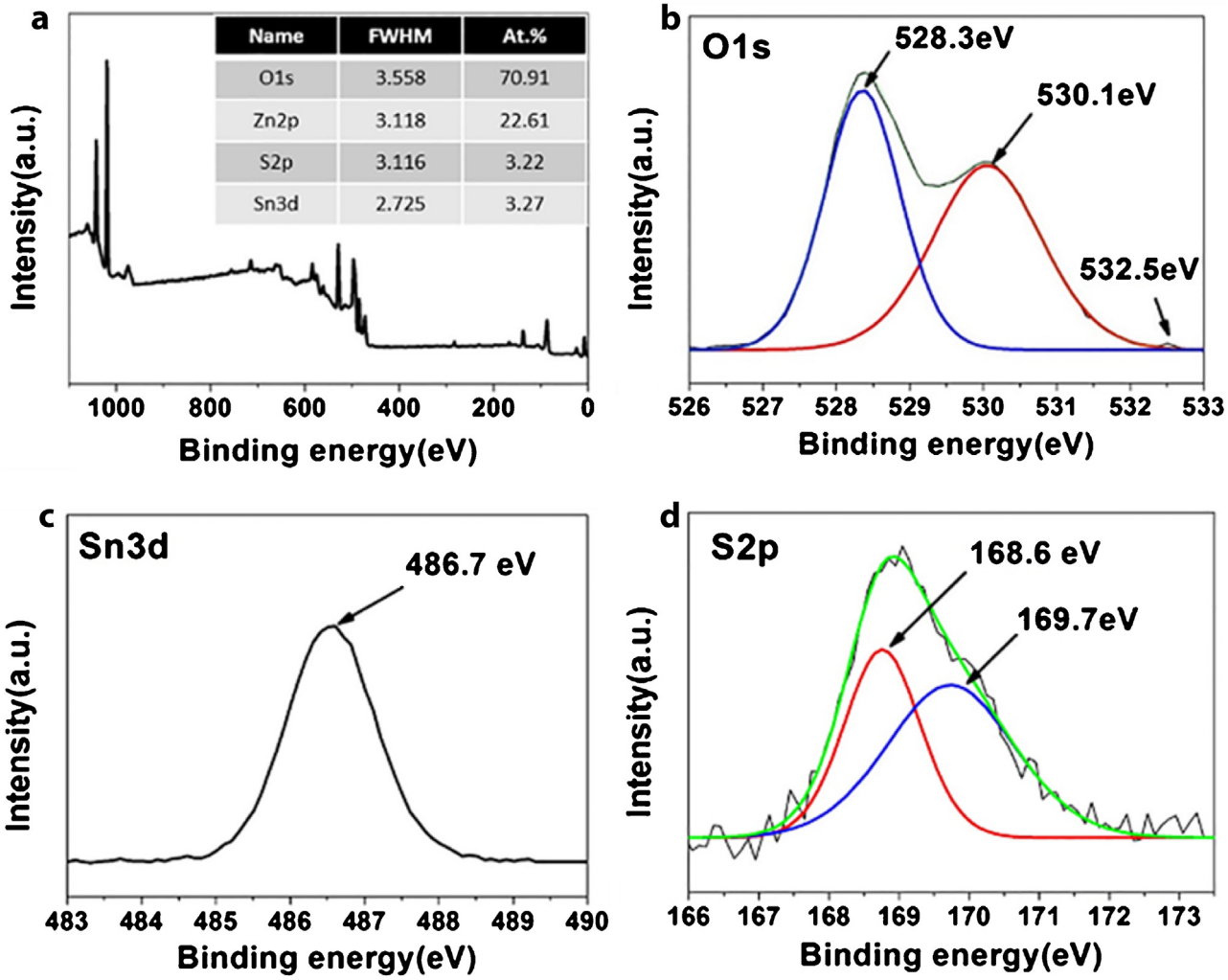


Fig. 2. XPS spectra of the ZnO-3%$SnSO_4$ sample (a) a full range spectrum and the corresponding surface elemental composition; (b) O1s; (c) Sn 3d; (d) S 2p.

The pure ZnO sample is composed of nanoparticles with average particle size of ~30 nm, as suggested by SEM (Fig. 1b). The particles are interconnected due to the sintering effect. The ZnO-3%$SnSO_4$ sample and ZnO particles are comparable in size, while the particles is connected into nanosheets, with lateral dimension of about 500 nm and thickness about 30 nm, i.e. the nanosheets are composed of one single layer of ZnO nanoparticle (Fig. 1c). Further increase the $SnSO_4$ concentration results in appearance of thin nanosheets together with the nanoparticles (Fig. S2a–f), clearly indicating that the connection of ZnO nanoparticles are induced by $SnSO_4$.

More detailed structural features of the ZnO-3%$SnSO_4$ sample are studied by HRTEM. Figs. 1e and S3 show that the ZnO nanoparticles are loaded on a thin layer in the nanosheets. The particles with dark contrast are Wurtzite ZnO while the thin layer with lighter contrast are composed of tetragonal SnO_2 nanocrystals, as suggested by the corresponding lattice images (Fig. 1f, g). Both the ZnO and SnO_2 phases are highly crystallized with well resolved lattice images, in agreement with the XRD results. Therefore, the SnO_2 thin layer serves as the template for the connection of the ZnO nanoparticles.

The interconnecting pattern of ZnO nanoparticles in the sensing layer strongly affects the electric conductivity behavior. The pure ZnO nanoparticles are simply connected with each other via the sintering neck. However, in the 2D nanosheets, considerable amount of the ZnO nanoparticles are not directly in contact with each other due to the presence of the SnO_2 phase. Instead, ZnO-SnO_2-ZnO heterojunctions are formed, as shown in Fig. 1h. A thin layer of SnO_2 is observed between two ZnO particles. In addition, a $Zn_3O(SO_4)_2$ nanocrystal is observed near the surface of one ZnO particle, which is formed by the reaction of the sulfate with the ZnO surface. Elemental mapping shows that the distributions of Zn, S and Sn are very homogeneous (Fig. 1i–l). Quantification yields Sn/Zn and Sn/S ratio close to that in the starting material.

The $SnSO_4$ modified ZnO nanoparticles during sintering is an interesting phenomenon which has not been reported previously. A possible mechanism is illustrated in Fig. 1d. $SnSO_4$ has a low melting point of 378 °C, below the sintering temperature (450 °C). Thus, before the $SnSO_4$ are oxidized according to Eq. (2), there will be a molten layer coated on the ZnO particles, which serves like a glue for the assembly of the ZnO particles. Then the oxidation reaction shown in Eq. (2) occurs, resulting in $Zn_3O(SO_4)_2$ nanocrystals attached to the ZnO surface and the SnO_2 nanosheet. The nanosheet morphologies are extensively observed for SnO_2. It has been shown that SnO_2 nanocrystals can aggregate into nanosheets under hydrothermal conditions [21–23]. The thin layers composed of SnO_2 nanocrystals may form via a similar mechanism, which serve as the template for the formation of the ZnO nanosheets. The $SnSO_4$ induced connection of ZnO nanoparticles is illustrated in Fig. 1d.

The ZnO-3%$SnSO_4$ sample is further studied by XPS (Fig. 2). In the Sn 3d5/2 spectrum (Fig. 2c), the peak at 486.7 eV is attributed

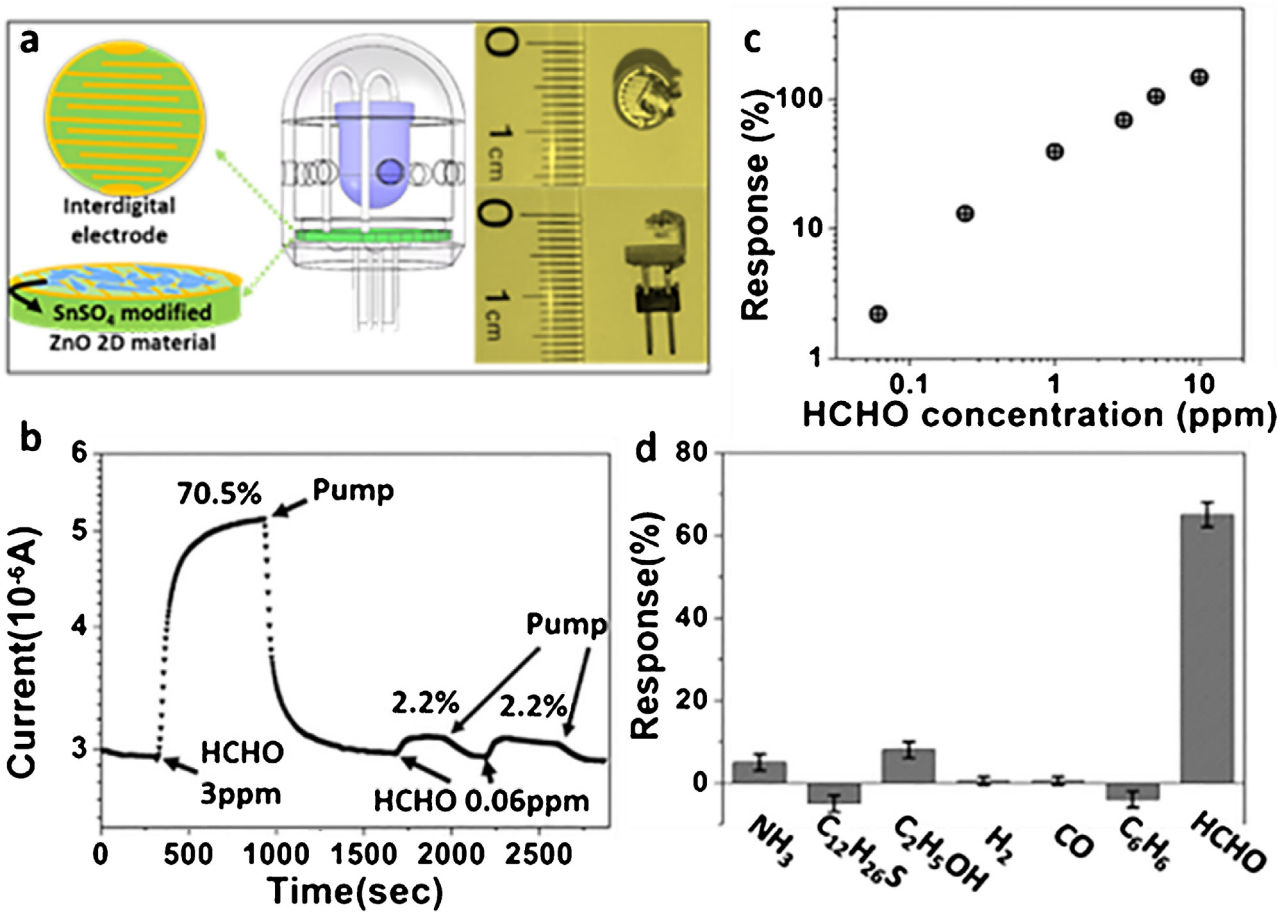


Fig. 3. (a) The schematic structure and photographs of the fabricated sensor; (b) The real-time response curve of the ZnO-3%$SnSO_4$ sample to 3 ppm and 0.06 ppm formaldehyde gas; (c) Responses of the ZnO-3%$SnSO_4$ sample to different concentration of formaldehyde; (d) Responses of the ZnO-3%$SnSO_4$ sample to 3 ppm various VOCs at room temperature.

to Sn (IV) [24]. The S 2p spectrum (Fig. 2d) can be deconvoluted into two peaks centered at 168.6 eV and 169.7 eV, which can be assigned to the sulfate species. Composition analysis from XPS gives Sn/Zn molar ratio of 1:7.2 and Sn/S molar ratio of 1.01:1 (Fig. 2a). The Sn/Zn ratio is much higher than the bulk value obtained from EDS while the Sn/S ratio remains close to that in $SnSO_4$, indicating surface enrichment of the tin and sulfate species. Therefore, both the XRD and XPS results reach excellent agreement for the surface chemical states of the elements, again confirming that $SnSO_4$ modification is a surface effect without notable Sn doping into the ZnO lattice. Moreover, the O 1s spectrum can be deconvoluted into two main components (Fig. 2b). The main peaks at binding energy of 528.3 eV and 530.1 eV are assigned to the lattice oxygen and defect oxygen in ZnO crystals and SnO_2, respectively [9]. The shoulder peak centered at 532.5 eV can be attributed to the chemisorbed oxygen species and hydroxyl groups, indicating an active surface of the $SnSO_4$ modified ZnO nanoparticles [25]. To evaluate the effect of $SnSO_4$ introduction on the energy gap of the ZnO samples, the optical properties of ZnO-3%$SnSO_4$ and ZnO samples are probed by UV–visible diffuse reflectance spectroscopy (Fig. S4). After calculating the band gap of the two samples, we can find that the band gap of ZnO-3%$SnSO_4$ is 2.15 eV, which is 0.15 eV smaller than that of the pure ZnO sample (2.3 eV).

The ZnO-x$SnSO_4$ samples are coated onto the PCB as described in the experimental section to fabricate the sensors. The sensor structure is schematically illustrated in Fig. 3a, which is very compact with a lateral dimension less than 2 cm even including the LED. The dynamic response characteristics of the ZnO-3%$SnSO_4$ sensor is shown in Fig. 3b. Rapid increase of the photocurrent is observed when the formaldehyde is injected. The sensor shows a positive response of 70.5% to 3 ppm formaldehyde in the steady state. The response time (defined as the time required for 90% amplitude of the photocurrent variation) is 100 s when formaldehyde is injected and 88 s for the recovery process when formaldehyde is removed, which is quite fast considering the fact that the sensor is operating near room temperature. In order to confirm the long term stability of the sensor, dynamic response of the sensor to 3 ppm formaldehyde is further tested after more than 3 months (Fig. S5). After repeated formaldehyde injection and removal, the response of the sensor to 3 ppm formaldehyde is maintained at around 68%, which is very close to the initial response of 70.5%. For very low formaldehyde concentration of only 0.06 ppm, a well-defined increase in photocurrent by 2.2% is observed for 0.06 ppm formaldehyde (Fig. 3b). The response is reversible and reproducible for repeated formaldehyde dosing. The response to formaldehyde concentration from 0.06 to 10 ppm is shown in Fig. 3c. Large difference in response is observed for formaldehyde concentration below 1 ppm. Thus, the ZnO-3%$SnSO_4$ sample is highly suited for quantitative analysis of low concentration of formaldehyde. Importantly, it has sufficient sensitivity for ultralow formaldehyde concentration to meet the strict WHO criterion (0.08 ppm). Such low formaldehyde concentration is very challenging for most MOS based sensors reported in literatures (Table S1) [26–36]. In addition to the advantages of the sensing material itself, the UV irradiation also plays an important role in promoting the sensing performance. Without UV irradiation, the ZnO-3%$SnSO_4$ shows no obvious response to 3 ppm

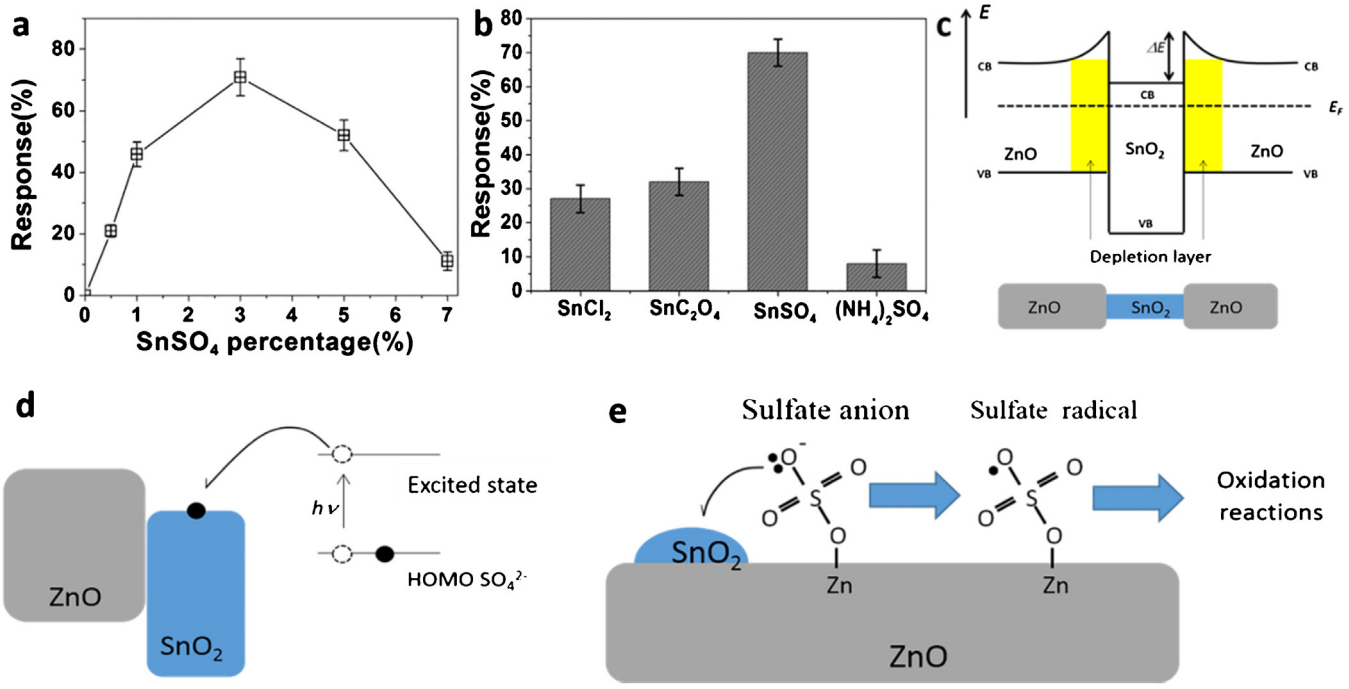


Fig. 4. (a) Responses to 3 ppm formaldehyde of sensors fabricated by ZnO-x$SnSO_4$ samples with different $SnSO_4$ percentage; (b) Responses to 3 ppm formaldehyde of sensors fabricated by ZnO nanoparticles modified by different reagents with concentration of 3 mol%; (c) Energy band diagram of the ZnO/SnO_2 heterojunction; (d) and (e) Schematic diagrams showing the formation mechanism of the sulfate radical under UV irradiation in the ZnO-x$SnSO_4$ sample.

formaldehyde (Fig. S6). Moreover, the background current drops to ~$5.6{\cdot}10^{-8}$ A, which is about two orders of magnitude smaller than that of the photocurrent.

The response to other commonly encountered volatile organic compounds (VOCs) in indoor environment, including ammonia, ethanol, 1-dodecanethiol and benzene, are also tested. As shown in Fig. 3d, the sensor shows very weak response to ammonia, ethanol, hydrogen and negative response to 1-dodecanethiol, benzene. Therefore, the sensor is able to sensitively distinguish formaldehyde from other commonly seen VOCs in indoor environment.

ZnO is an n-type semiconductor due to the intrinsic oxygen deficiency. The oxygen in the air will be adsorbed on the ZnO surface, which serve as electron traps. The surface chemisorbed oxygen O_{ads} gives rise to the shoulder at 532.5 eV in the O2p XPS spectrum (Fig. 2c). Reducing the density of O_{ads} will release the trapped electrons and promote the electric conductivity. Thus, the electric conductivity of ZnO will increase for reducing gas molecules such as formaldehyde and ethanol. On the other hand, gases that may result in higher O_{ads} density will lower the electric conductivity, such as NOx [37,38].

Light irradiation is used to increase the carrier density and also promote the surface redox reactions to give better sensitivity. In fact, thermal activation is more frequently used in MOS gas sensors. The operation temperature for most MOS based gas sensors is 200 °C or above [39,40]. The high temperature accelerates the surface redox kinetics and leads to faster response. However, many reductive gases can be reduced at high temperature. As a result, thermally activated MOS sensors exhibit response to a large variety of reductive gases in addition to formaldehyde, including ethanol, 1-dodecanethiol and ammonia [23,41,42]. On the other hand, the light activation approach significantly improves the selectivity, as shown in the paper. Formaldehyde is a strong reducing agent and very easy to be oxidized. The π electrons in the carbonyl are also easier to activate by UV irradiation, resulting in higher reactivity in the surface photochemical redox process compared to other reductive gases. Thus, the light activation approach is more suited for specific formaldehyde detection.

Proper $SnSO_4$ percentage in the ZnO-x$SnSO_4$ samples is critical for the detection sensitivity. Fig. 4a shows the response to 3 ppm formaldehyde of different ZnO-x$SnSO_4$ samples (x = 0–7%). The pristine ZnO nanoparticles exhibit almost no response. Low level of $SnSO_4$ modification drastically improves the sensitivity. The optimal $SnSO_4$ percentage is 3%, holding the highest response of 70.5%. The response then decreases with further increase of the $SnSO_4$ percentage. Moreover, the anion in the tin salt also strongly affects the sensitivity. As shown in Fig. 4b, the response is reduced by almost 43% and 38% at the same formaldehyde concentration when replacing $SnSO_4$ with $SnCl_2$ and SnC_2O_4, respectively. In addition, modification using $(NH_4)_2SO_4$ only, i.e. no metal cations, shows only little improvement in sensitivity. The SnO_2 phase and the sulfate anion exhibit clear synergic effect in promoting the formaldehyde sensitivity.

$SnSO_4$ is converted into SnO_2 and $Zn_3O(SO_4)_2$ after calcination. SnO_2 has been extensively studied for gas sensing, which has a wider band gap of 3.6 eV, compared to 3.2 eV of ZnO. The conduction band of SnO_2 is lower than that of ZnO. The electron affinity of ZnO and SnO_2 is 4.5 and 4.9 eV, respectively. The Fermi level of ZnO was higher than that of SnO_2 due to its smaller work function. Accordingly, when ZnO and SnO_2 formed a heterojunction, the electron will transfer from ZnO to SnO_2, forming an electron depletion layer on the ZnO side and an electron accumulation layer on the SnO_2 side [3,24,43]. There is an energy barrier for electron transport, as shown by the term ΔE in Fig. 4c.

The thickness of the depletion layer and also the height of the energy barrier ΔE can be modulated by the surface adsorbed oxygen species (O_{ads}). When O_{ads} is removed through the chemical reaction with formaldehyde, ΔE is reduced and the thickness of the depletion layer becomes thinner. As a result, the electrons trapped in the heterojunction region can be released, resulting in higher electric conductivity [44,45]. Briefly, the ZnO/SnO_2 heterojunction acts as a lever in electron transfer through which electron transfer is facilitated or restrained depending on the surface adsorbed oxygen species, which amplify the electric conductivity change when the surface adsorbed species changes [44].

With this picture, the effect of $SnSO_4$ concentration can be easily understood. For pristine ZnO nanoparticles, the particles are directly attached to each other via the sintering neck. The effect of the surface adsorbed oxygen species can only trap some electrons near the surface, forming a surface depletion layer. Change of the surface adsorbed species only shows weak modulation effect on the electric conductivity. With the increasing concentration of $SnSO_4$, the ZnO nanoparticles are connected with each other via the ZnO/SnO_2 heterojunctions. The surface adsorbed oxygen species strongly affects the thickness of the depletion layer and the height of the energy barrier, resulting in very high sensitivity to the formaldehyde. When the $SnSO_4$ concentration further increases, nanosheets from the excessive SnO_2 gradually appear, as shown in the SEM images (Fig. S2). The SnO_2 nanosheets providing alternative electric conductivity pathways and the modulation effect of the surface adsorbed oxygen species on the electric conductivity of the sensing layer becomes weak again.

The effect of the surface sulfate anion is more intriguing. As far as we know, there has been little discussion on this issue. Here we propose that the surface SO_4^{2-} may generate sulfate radical anions $SO_4^{-\bullet}$ under UV irradiation by exciting one electron from SO_4^{2-}. The excited electron can be further transferred into the conduction band of SnO_2 due to the electron trap nature of SnO_2 in the ZnO/SnO_2 heterojunction. The sulfate radical anions $SO_4^{-\bullet}$ is a very strong oxidation agent with very high oxidative potential of 2.4 V, which has been extensively used in removing the organic pollutants in water [46,47]. $SO_4^{-\bullet}$ can react with the surface adsorbed oxygen or hydroxyl groups, yielding highly reactive oxidative species such as $O_2^{-\bullet}$, $O^{-\bullet}$ and $^{\bullet}OH$, and accelerate the formaldehyde oxidation process (Fig. 4d, e) [46,47]. Further investigation is ongoing to confirm the above mechanism.

It should be noted that the nanosheet structure of the ZnO-3%$SnSO_4$ sample is also beneficial to enhance the sensitivity. It provides high density of interaction sites while maintains good electric contact between the nanoparticles. Thus, the simple surface modification using $SnSO_4$ results in several key features advantageous for formaldehyde detection simultaneously, including forming the ZnO/SnO_2 heterojunction, sulfate modification and inducing connection of ZnO nanoparticles. The ZnO nanoparticle preparation and surface modification involves only low cost materials and easy and scalable operations, which is highly promising for practical application in compact portable formaldehyde sensors.

4. Conclusions

ZnO nanoparticles with surface modification by $SnSO_4$ can form nanosheet structure after sintering at 450 °C in air. The nanosheets are composed of a single layer of ZnO nanoparticles loaded on thin layers composed of SnO_2 nanocrystals. Formation of the ZnO/SnO_2 heterojunctions and the surface sulfate significantly promote the formaldehyde sensing performance. The ZnO-3%$SnSO_4$ sensor is able to detect ultralow concentration of formaldehyde (0.06 ppm) with excellent selectivity from commonly occurred indoor VOCs. The high sensing performance achieved by widely available ZnO nanoparticles with simple modification is very encouraging to develop high performance, low cost, compact formaldehyde sensors.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.snb.2017.05.023.

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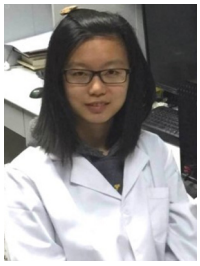
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Biographies



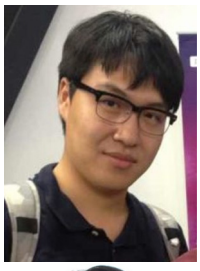
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